UNIT IV (Fuzzy Membership, Rules)

Lecture #1 Membership Function

A membership function defines fuzziness in the fuzzy set. It may be discrete or continuous. The membership function is generally represented graphical form. Standard shapes of membership functions are maintained over the years. Fuzzy membership functions are determined in practical problems with opinion of experts.

As we know, a fuzzy set $\tilde{A}$ in the universe of discourse X can be as

$$\tilde{A} = \{(x, \mu_{\tilde{A}}(x)) : x \in \tilde{A}\}$$

Where, $\mu_{\tilde{A}}(x)$ is called membership function. The membership function, $\mu_{\tilde{A}}(x)$ maps X on the interval $[0 \quad 1]$

$$\mu_{\tilde{A}}(x) : X \longrightarrow [0 \quad 1]$$

The range of the membership function is subset of the non – negative real numbers whose supremum is finite.

Features of the membership function

Three main basic features of membership function are as shown in figure 1 a

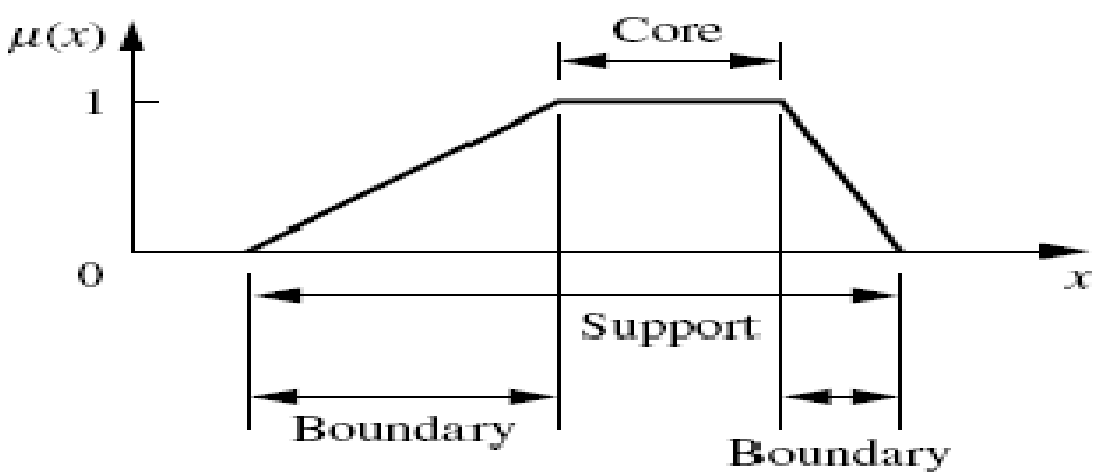


Figure 1 a Feature of membership function

1 **CORE** The *core* of a membership function for some fuzzy set $\tilde{A}$ is defined as that region of the universe that is characterized by complete and full membership in the set$\tilde{A}$. That is, the core comprises those elements *x* of the universe **X**

such that $\mu_{\tilde{A}}(x) = 1$

2 **Support** .The *support* of a membership function for some fuzzy set $\breve{A}$ is defined as that region of the universe that is characterized by nonzero membership in the set$\breve{A}$. That is, the support comprises those elements *x* of the universe **X**

such that $\mu_{\tilde{A}}(x) > 0$.

3. Boundary. The *boundaries* of a membership function for some fuzzy set *A*are defined as that region of the universe containing elements that have a nonzero membership but not complete membership

$0 < \mu_{\tilde{A}}(x) < 0$.

4. Normal and Subnormal Fuzzy Sets

A *normal* fuzzy set is one whose membership function has at least one element **x** in ***X*** the universe whose membership value is unity. Otherwise it will be subnormal fuzzy set as shown in figure 1(b) and figure 1(c).

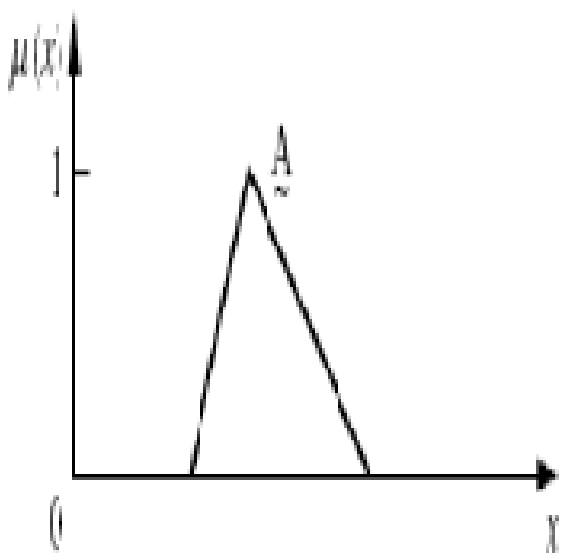


Figure 1(b) Normal Fuzzy set

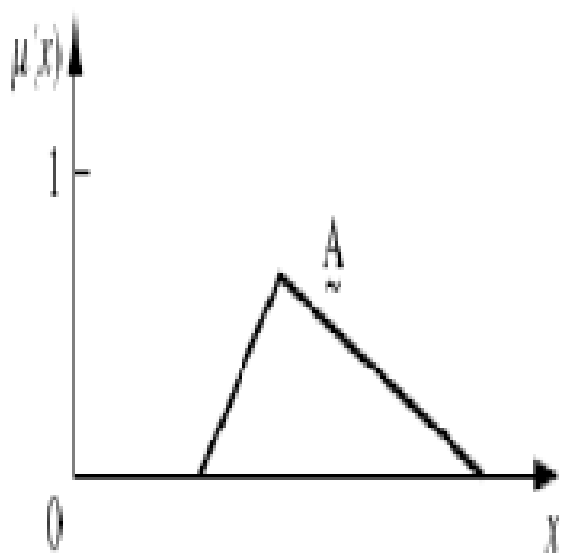


Figure 1(c) Subnormal Fuzzy set

Convex fuzzy set

A *convex* fuzzy set is described by a membership function whose membership values are strictly monotonically increasing, or strictly monotonically decreasing, or strictly monotonically increasing then strictly monotonically decreasing with increasing values for elements in the universe.

(Said another way, if, for any elements x, y, and z in a fuzzy set A, the relation $x < y < z$ implies that

$$\mu_{\tilde{A}}(y) = \min\lfloor \mu_{\tilde{A}}(x), \mu_{\tilde{A}}(z) \rfloor$$

Otherwise set will be non-convex fuzzy set as shown in figure 1(d) and figure 1(e).

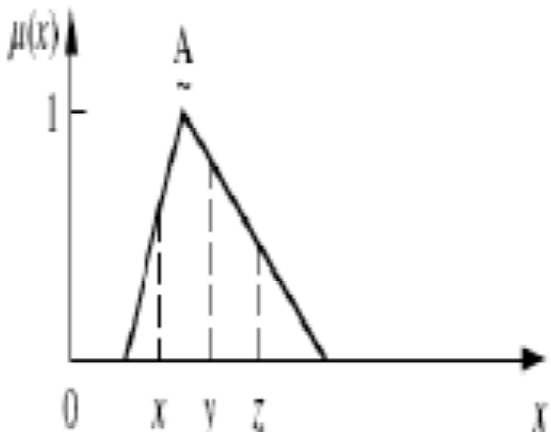


Figure 1 (d) convex fuzzy set

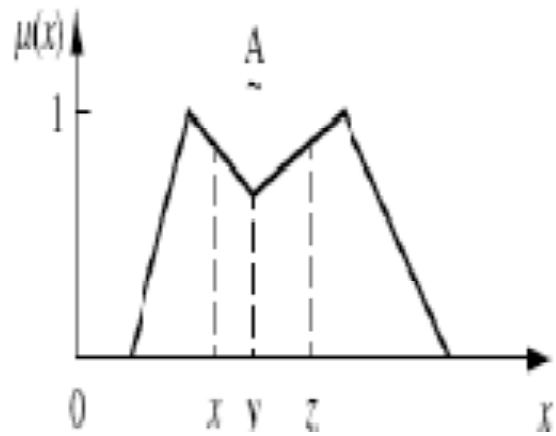


Figure 1 (e) Norn – convex fuzzy set

Crossover Point

The crossover points of a membership function are defined as the elements in the universe X for which a particular fuzzy set $\tilde{A}$ has values equal to 0.5, i.e., for which

$\mu_{\tilde{A}}(x) = 0.5$.

Height of a Fuzzy Set

The height of a fuzzy set $\tilde{A}$ is the maximum value of the membership function, i.e.,

height $(\tilde{A}) = \max(\mu_{\tilde{A}}(x))$

For subnormal fuzzy set height $(\tilde{A}) < 1$

Lecture # 2 Inferences in Fuzzy Logic, Fuzzy IF-THEN Rules

In crisp logic, the truth values acquired by proposition or predicate are 2-valued, namely *True*, *False* which may be treated numerically equivalent to (0, 1). However, in *fuzzy* logic, the truth values are multivalued such as *absolutely true, very true, partly true, absolutely false* and so on numerically equivalent to (0-1).

Fuzzy Propositions

A fuzzy *proposition* is a statement which acquires a fuzzy truth value. Thus, given $\tilde{P}$to be fuzzy proposition, T ($\tilde{P}$) represents the truth value (0-1) attached to$\tilde{P}$. In its simplest form, fuzzy propositions are associated with fuzzy sets. The fuzzy membership value associated with the fuzzy set Ã for P is treated as the fuzzy truth value T ($\tilde{P}$).

i.e. T ($\tilde{P}$) =$\mu_{\tilde{A}}$ (x) where 0≤ $\mu\tilde{A}$(x) ≤1

Example

$\tilde{P}$: Ram is honest.
T ($\tilde{P}$) = 0.8 if P is partly true
T ($\tilde{P}$) = 1 if P is absolutely true

Fuzzy Connectives

Fuzzy logic, similar to crisp logic, support following connectives;

(i) Negation ; -
(ii) Disjunction; v
(iii) Conjunction; ^
(iv) Implication; =>

Let $\tilde{P}$ and $\tilde{Q}$ are two fuzzy propositions with relative truth values are T (P) and T (Q) then connectives can be defined as

Symbol	Connective	Usage	Definition
-	Negation	$\tilde{P}$	**1-T($\tilde{P}$)**
V	Disjunction	$\tilde{p} \bigvee \tilde{Q}$	**max(T($\tilde{P}$), T($\tilde{Q}$))**
^	Conjunction	$\tilde{P} \bigwedge \tilde{Q}$	**min(T($\tilde{P}$), T($\tilde{Q}$))**
=>	Implication	$\tilde{P} => \tilde{Q}$	**Max(1-T($\tilde{P}$), T($\tilde{Q}$))**

$\tilde{P}$ and $\tilde{Q}$ are related by '=>' operator are known as antecedent and consequent respectively. Also, just as in crisp logic, here too, '=>' represent the **IF – THEN** statement as

IF x is $\tilde{A}$ THEN y is $\tilde{B}$, and is equivalent to

$$\tilde{R} = (\tilde{A} \times \tilde{B}) \cup (\overline{\tilde{A}} \times Y)$$

The *membership function of* $\tilde{R}$ *is given by*

$$\boxed{\mu_{\tilde{R}}(x, y) = \max(\min(\mu_{\tilde{A}}(x), \mu_{\tilde{B}}(y)), 1-\mu_{\tilde{A}}(x))}$$

Also for compound implication **IF x is A THEN y is B ELSE y is C** the relation R is equivalent to

$$\tilde{R} = \tilde{R} = (\tilde{A} \times \tilde{B}) \cup (\overline{\tilde{A}} \times \tilde{C})$$

The membership function of R is given by

$$\boxed{\mu_{\tilde{R}}(x, y) = \max(\min(\mu_{\tilde{A}}(x), \mu_{\tilde{B}}(y)), \min(1-\mu_{\tilde{A}}(x), \mu_{\tilde{C}}(y)))}$$

Example

$\tilde{P}$: Mary is efficient, T ($\widetilde{P}$) =0.8

$\tilde{Q}$: Ram is efficient, T ($\tilde{Q}$) =0.65

(i) $\tilde{P}$: =Mary is not efficient.

T ($\tilde{P}$) =1-T ($\tilde{P}$) = 1-0.8 =0.2

(ii) $\tilde{P} \wedge \tilde{Q}$: Mary is efficient and so is Ram

T ($\tilde{P} \wedge \tilde{Q}$) = min (T ($\tilde{P}$), T ($\tilde{Q}$)) = min (0.8, 0.65) = 0.65

(iii) $\tilde{P} \vee \tilde{Q}$: Either Mary or Ram is efficient

T ($\tilde{P} \vee \tilde{Q}$) = max (T ($\tilde{P}$), T ($\tilde{Q}$)) = (0.8, 0.65) = 0.8

(iv) $\tilde{P}$=>$\tilde{Q}$: If Marry is efficient then so Ram is

T ($\tilde{P}$=>$\tilde{Q}$) =max ((1-T ($\tilde{P}$), T ($\tilde{Q}$)) =max (0.2, 0.65) =0.65

Example (To understand IF-THEN rule)

Let X= {a, b, c, d}, Y= {1, 2, 3, 4}

and $\tilde{A}$ = {(a, 0) (b, 0.8) (c, 0.6) (d, 1)}

$\tilde{B}$ = {(1, 0.2) (2, 1) (3, 0.8) (4, 0)}

$\tilde{C}$= {(1, 0) (2, 0.4) (3, 1) (4, 0.8)}

Determine the implication relations

(i) ***IF x is $\tilde{A}$ THEN y is $\tilde{B}$***

(ii) ***IF x is $\tilde{A}$ THEN y is $\tilde{B}$ ELSE y is $\tilde{C}$***

<u>Solution</u>

To determine Part (i)

Compute $\tilde{R}$= **($\tilde{A}$ ×$\tilde{B}$) ∪ ($\bar{\tilde{A}}$ ×Y) where**

The *membership function of $\tilde{R}$ is given by*

$$\boxed{\mu_{\tilde{R}}(x, y) = \max(\min(\mu_{\tilde{A}}(x), \mu_{\tilde{B}}(y)), 1-\mu_{\tilde{A}}(x))}$$

where $\mu_R(x, y) = \max(\min(\mu_A(x), \mu_B(y)), 1-\mu_A(x))$

$$\tilde{A} \times \tilde{B} = \begin{array}{c} \\ a \\ b \\ c \\ d \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0.2 & 0.8 & 0.8 & 0 \\ 0.2 & 06 & 0.6 & 0 \\ 0.2 & 1 & 0.8 & 0 \end{pmatrix} \end{array}$$

and $\overline{\widetilde{A}} \times \mathbf{Y}$=

$$\begin{array}{c} \\ a \\ b \\ c \\ d \end{array} \begin{array}{c} \begin{array}{cccc} \mathbf{1} & \mathbf{2} & \mathbf{3} & \mathbf{4} \end{array} \\ \begin{pmatrix} \mathbf{1} & \mathbf{1} & \mathbf{1} & \mathbf{1} \\ \mathbf{0.2} & \mathbf{0.2} & \mathbf{0.2} & \mathbf{0.2} \\ \mathbf{0.4} & \mathbf{0.4} & \mathbf{0.4} & \mathbf{0.4} \\ 0 & 0 & 0 & 0 \end{pmatrix} \end{array}$$

Here Y is Universe of discourse could be viewed as **{(1, 1)(2,1)(3,1)(4,1)}** a fuzzy set of all whose elements x have μ(x)=1.

Therefore $\tilde{R}$=max (**($\widetilde{A} \times \widetilde{B}$) ∪ ($\overline{\widetilde{A}} \times$Y)**

$$\tilde{R}= \begin{array}{c} \\ a \\ b \\ c \\ d \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0.2 & 0.8 & 0.8 & 0.2 \\ 0.4 & 0.6 & 0.6 & 0.6 \\ 0 & 0 & 0 & 0 \end{pmatrix} \end{array}$$

Which represents IF x is $\tilde{A}$ THEN y is $\tilde{B}$

To determine Part II

Compute Also for compound implication **IF x is $\widetilde{A}$ THEN y is $\widetilde{B}$ ELSE y is $\widetilde{C}$** the relation $\tilde{R}$ is equivalent to

$\widetilde{R}$ = **($\widetilde{A} \times \widetilde{B}$) ∪ ($\overline{\widetilde{A}} \times \widetilde{C}$)**

The membership function of R is given by

μ$\tilde{R}$ (x, y) =max (min (μ$\widetilde{A}$ (x), μ$\widetilde{B}$ (y)), min ((1-μ$\widetilde{A}$ (x), μ$\widetilde{C}$ (y)))

$$\widetilde{A} \times \widetilde{B} = \begin{array}{c} \\ a \\ b \\ c \\ d \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0.2 & 0.8 & 0.8 & 0 \\ 0.2 & 06 & 0.6 & 0 \\ 0.2 & 1 & 0.8 & 0 \end{pmatrix} \end{array}$$

$$\overline{\widetilde{A}} \times \widetilde{C} = \begin{array}{c} \\ a \\ b \\ c \\ d \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{pmatrix} 0 & 0.4 & 1 & 0.8 \\ 0 & 0.2 & 0.2 & 0.2 \\ 0 & 0.4 & 0.4 & 0.4 \\ 0 & 0 & 0 & 0 \end{pmatrix} \end{array}$$

Therefore $\widetilde{R}$ = max (**($\widetilde{A} \times \widetilde{B}$), ($\overline{\widetilde{A}} \times \widetilde{C}$)**) gives

$$\tilde{R}= \begin{array}{c} \\ a \\ b \\ c \\ d \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{pmatrix} 0 & 0.4 & 1 & 0.8 \\ 0.2 & 0.8 & 0.8 & 0.2 \\ 0.2 & 0.6 & 0.6 & 0.4 \\ 0.2 & 1 & 0 & 0 \end{pmatrix} \end{array}$$

The above relation represents IF x is $\tilde{A}$ is Then y is $\tilde{B}$ ELSE y is $\tilde{C}$

Fuzzy Quantifiers (Not Important from Examination point of View –for Knowledge You may skip this topic)

Just as in crisp logic where predicates are quantified by quantifiers, fuzzy logic propositions are also quantified by fuzzy quantifiers. *The scope fuzzy propositions can be extended using fuzzy quantifier.* There are two classes of fuzzy quantifiers such as

(i) Absolute quantifiers

(ii) Relative quantifiers.

While absolute quantifiers are defined over *R* *(Set of real numbers)*

*Example "***about to 10, much more than 100, at least about 5"**

Relative quantifiers are defined over [0-1].

Ex. "Almost All", "About Half", "most"

Propositions involved in absolute quantifier:

P; There Q I's in I such that V (i) in F

- Q is an absolute quantifier

-i am an individual from a given set I

V (i) is a variable associated to individual I that take value from universe E.

F is a fuzzy set defined over universe E.

Ex. ***"There is <u>about 10 students in a given class</u> whose <u>fluency in English V (i)</u> is <u>high</u>',***

Q="about 10" I's student I class, V(i) Fluency in English of student I F is a fuzzy set that represents a high degree of fluency in English.

It can be converted to simplest form

P': There are Q Z's in I

Z is a fuzzy set defined as follows

Z is a fuzzy set defined as follows:

□*z* (*i*) □.□*F* (*V* (*i*)) *i*□*I*

• Thus, the former proposition is replaced with:

"There are <u>about 10 high-fluency English speaking students</u> in a <u>given class</u>"

To calculate the truth value of *p*' we need to calculate the cardinality of fuzzy set *Z*

- The cardinality of *Z* can be calculated as

follows: $|Z|=\Sigma_{i€I}\mu_Z(i)= \Sigma_{i€I}\mu_F V(i))$

- Then, we calculate the truth value T (*p'*)

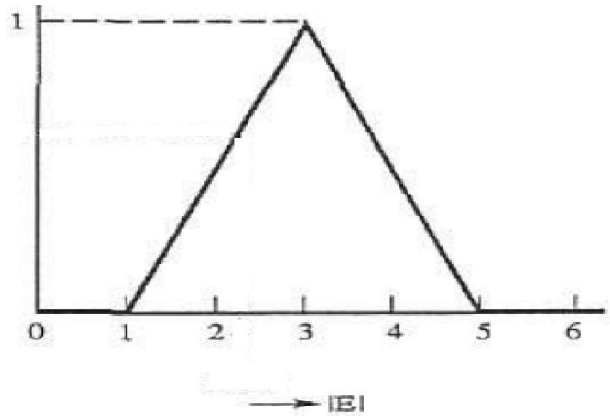
using the membership $T(p')=\mu_Q(|Z|)$

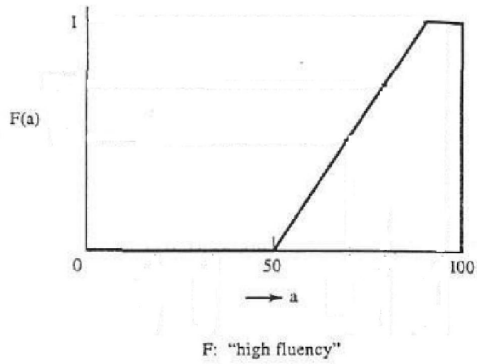
Example "*There are about 3 students in I whose fluency in English V (i) is high"*

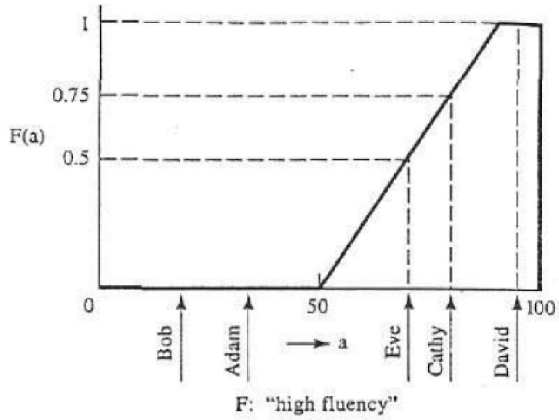
• *I* = {Adam, Bob, Cathy, David, Eve}

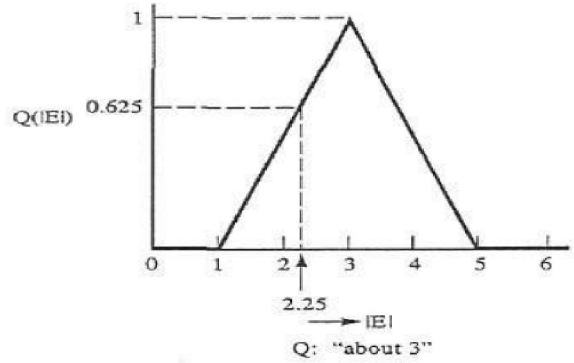
• *V* = Degree of fluency in English

– *V*(Adam)= 35, *V*(Bob)=20, *V*(Cathy)=80, *V*(David)=99, *V*(Eve)=70









Type #1 fuzzy quantifiers may also appear in more complex propositions:

p: There are *Q i*'s in *I* such that *V1* (*i*) is *F1* and

V2 (*i*) is *F2*

• Example:

"There are at least about two students in the class whose fluency in English is high and are young"

Alternative format for propositions of the former type:

p': There is *Q Y* in *I*,

Y = T-norm (ìF1 (*V1 (i*)), ìF2 (*V2 (i*)))

• **Example**:

p': "There are at least about two high-fluency English-speaking and young students in the class"

• Obviously *Y* is the set of "English-speaking and young students"

Exercise (Homework)

• Calculate the truth value of the proposition:

"***There are at least about 3 students in the class whose fluency in English is high and are young"***

• Use the data in slide #7 (students) for: *I, V* (*i*), ìQ and ìF

• Use the following T-norm: T(x, y) = min(x, y)

• The ages for the different students are: – Age(Adam)=23, Age(Bob)=35, Age(Cathy)=46, Age(David)=54, Age(Eve)=25

<u>Exercise (Homework)</u>

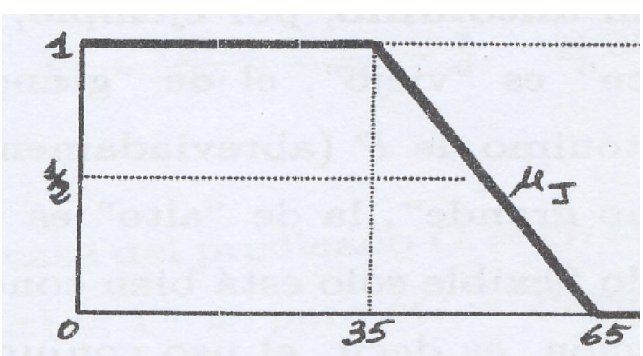
• Calculate the truth value of the proposition: "There are at least about 3 students in the class whose fluency in English is high and are young"

• Use the data in slide #7 (students) for: *I, V* (*i*), ìQ and ìF

• Use the following T-norm: T(x, y) = min(x, y)

• The ages for the different students are:

– Age(Adam)=23, Age(Bob)=35, Age(Cathy)=46,– Age(David)=54, Age(Eve)=25



It is also possible to have propositions involving relative quantifiers:

p: Among i's in *I* such that *V1* (*i*) is *F*1 there are *Q i*'s
in *I* such that *V*2 (*i*) is *F*2

• Example:

"Among students in a given class that are young, there are almost all whose fluency in English is high"

This kind of propositions can be rewritten as:

p': *Q Y*'s are *Z*'s

Y = ìF1 (*V*F1 (*i*)), *Z* = ìF2 (*V*F2 (*i*))

• Example:

"Almost all young students in a given class are students whose fluency in English is high"

• How do we calculate the truth value of such propositions?

We can rewrite *p'* as *p''*: *W* is *Q*

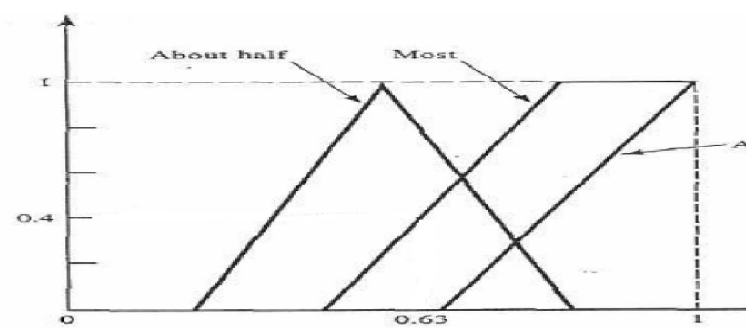
• *W* is the degree of subsethood of *Y* in *Z*

– How do we calculate *W*?

– $$W = \frac{|Z \cap Y|}{|Z|} = \frac{\sum_{i \in I} \min\left(\mu_{F_1}(V_1(i)), \mu_{F_2}(V_2(i))\right)}{\sum_{i \in I} \mu_{F_1}(V_1(i))}$$

Once we have obtained *W*, we can easily calculate *T* (p)

– $$T(p) = \mu_Q(W)$$



Lecture #3 Fuzzy Inferences

Fuzzy inference also referred to as approximate reasoning refers to computational procedures used for evaluating linguistic descriptions. The two important inferring procedures are

(i) GMP (Generalized Modes Ponens

(ii) GMT(Generalized Modes Tollens)

GMP is formally stated as,

$$\frac{\begin{array}{c} \textbf{\textit{IF x is } } \widetilde{\boldsymbol{A}} \textbf{\textit{ THEN y is } } \widetilde{\boldsymbol{B}} \\ \boldsymbol{x \text{ is } \widetilde{A}'} \end{array}}{\boldsymbol{y \text{ is } \widetilde{B}'}}$$

Here $\widetilde{\mathbf{A}}$, $\widetilde{\mathbf{B}}$ $\widetilde{\mathbf{A}}$', $\widetilde{\mathbf{B}}$ 'are fuzzy terms. Every fuzzy linguistic statement above the line is analytically known and what is below is analytically unknown.

To compute the membership function of $\widetilde{B}$' the max-min composition of fuzzy set $\widetilde{\boldsymbol{A}}$' with $\breve{\boldsymbol{R}}$**(x, y)** which is the known implication relation (IF-THEN) is used. That is,

$$\boxed{\widetilde{\mathbf{B}}' = \widetilde{\mathbf{A}}' \circ \widetilde{\mathbf{R}}(\mathbf{x}, \mathbf{y})}$$

In term of membership function

$$\boxed{\mu_{\widetilde{B}'}(\mathbf{y}) = \max\left(\min\left(\mu_{\widetilde{A}'}(\mathbf{x}), \mu_{\breve{R}}(\mathbf{x}, \mathbf{y})\right)\right)}$$

Where $\mu_{\widetilde{A}'}$ **(x)** is the membership function of $\tilde{A}$' $\mu_{\breve{R}}$**(x, y)** is the membership function of the implication relation and $\mu_{\widetilde{B}'}$ **(y)** is the membership function of $\tilde{B}$'

On the other hand, GMT has the form

IF x is $\tilde{A}$ THEN y is $\tilde{B}$

$$\frac{y \text{ is } \widetilde{B}'}{x \text{ is } \tilde{A}'}$$

To compute the membership function of $\widetilde{A}$' the max-min composition of fuzzy set $\widetilde{B}$' with $\breve{R}$(x, y) which is the known implication relation (IF-THEN) is used. That is,

$$\boxed{A' = \tilde{B}' \text{o} \tilde{R}(x, y)}$$

In term of membership function

$\mu\widetilde{A}'(x) = \max(\min(\mu\widetilde{B}'(y), \mu\widetilde{R}(x, y)))$

Where $\mu_{\widetilde{A}'}$ **(x)** is the membership function of $\widetilde{A}$' $\mu_{\breve{R}}$**(x, y)** is the membership function of the implication relation and $\mu_{\widetilde{B}'}$ **(y)** is the membership function of $\tilde{B}$'

$$\boxed{\mu_{\widetilde{A}'}(x) = \max(\min(\mu_{\widetilde{B}'}(y), \mu_{\breve{R}}(x, y)))}$$

Solved Example on Fuzzy Inference

Apply the fuzzy Modes Ponens rule to deduce Rotation is quite slow

(i) If the temperature is high then rotation is slow.

(ii) The temperature is very high

Let $\widetilde{H}$(High), $\tilde{V}$H (Very High) $\tilde{S}$(Slow) and $\tilde{Q}$S (Quite Slow) indicate the associated fuzzy sets as follows

For X= {30, 40, 50, 60, 70, 80, 90, 100}, the set of temperatures and Y= {10, 20, 30, 40, 50, 60} the set of rotations per minute

$\widetilde{H}$= {(70, 1) (80, 1) (90, 0.3)}

$\tilde{V}$H = {(90, 0.9) (100, 1)}

$\tilde{Q}$S = {(10, 1) (20, 0.8)}

$\tilde{S}$= ((30, 0.8) (40, 1) (50, 0.6)}

To derive $\tilde{R}$(x, y) representing the implication relation (i), we need to compute

$\widetilde{R}(x, y) = \max(\widetilde{H} \times \tilde{S}, \overline{\widetilde{H}} \times Y)$

$$\tilde{H}\times\tilde{S} = \begin{array}{c} \\ 30 \\ 40 \\ 50 \\ 60 \\ 70 \\ 80 \\ 90 \\ 100 \end{array} \begin{array}{c} \begin{array}{cccccc} 10 & 20 & 30 & 40 & 50 & 60 \end{array} \\ \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.8 & 1 & 0.6 & 0 \\ 0 & 0 & 0.8 & 1 & 0.6 & 0 \\ 0 & 0 & 0.3 & 0.3 & 0.3 & \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \end{array}$$

$$\overline{\tilde{H}}\times Y = \begin{array}{c} \\ 30 \\ 40 \\ 50 \\ 60 \\ 70 \\ 80 \\ 90 \\ 100 \end{array} \begin{array}{c} \begin{array}{cccccc} 10 & 20 & 30 & 40 & 50 & 60 \end{array} \\ \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0.7 & 0.7 & 0.7 & 0.7 & 0.7 & 0.7 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix} \end{array}$$

$$\tilde{R}(x, y) = \max(\tilde{H}\times\tilde{S}, \overline{\tilde{H}}\times Y) = \begin{array}{c} \\ 30 \\ 40 \\ 50 \\ 60 \\ 70 \\ 80 \\ 90 \\ 100 \end{array} \begin{array}{c} \begin{array}{cccccc} 10 & 20 & 30 & 40 & 50 & 60 \end{array} \\ \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0.8 & 1 & 0.6 & 0 \\ 0 & 0 & 0.8 & 1 & 0.6 & 0 \\ 0.7 & 0.7 & 0.7 & 0.7 & 0.7 & 0.7 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix} \end{array}$$

To deduce Rotation is quite slow we make use of the composition

$\tilde{Q}S = \tilde{H} o \tilde{R}(x, y)$ (set $\tilde{H}$ has membership vales for elements 90, and 100 only that 0.9 and 1 respectively. Membership values for remaining elements are 0)

$= [0\ 0\ 0\ 0\ 0\ 0\ 0.9\ 1] \times \tilde{R}(x, y) = [1\ 1\ 1\ 1\ 1\ 1]$

NOTE: ***(Apply max-min composition on matrices $\tilde{H}$ and $\tilde{R}(x, y)$)***

Lecture # 4 Fuzzification & Defuzzification

Fuzzy Rule Based Systems

Fuzzy linguistic descriptions are formal representations of systems made through fuzzy IF-THEN rules. They encode knowledge about a system in statement of the form-

IF (a set of conditions) are satisfied THEN (a set of consequent) can be inferred,

Fuzzy IF-then rule is coded in the form

IF(x_1 is $\tilde{A}_1$, x_2 is $\tilde{A}_2$ …..x_n is $\tilde{A}_n$) THEN (y_1 is $\tilde{B}_1$, y_2 is $\tilde{B}_2$….y_n is $\tilde{B}_n$)

where linguistic variables x_i, y_j take the values of fuzzy sets $\tilde{A}$i and $\check{B}_j$ respectively.

Where i, j =1,2, ….,n

Example for fuzzy Rule Base

If there is heavy rain and strong winds

Then there must be severe flood warning.

Here **heavy**, **strong**, and **severe** are fuzzy sets qualifying variables **rain, wind, and flood warning** respectively.

A collection of rules referring to a particular system is known as a *fuzzy rule base*. If the conclusion **C** to be drawn from a rule base **R** is the conjunction of all the individual consequents C_i of the each rule. Then

$$C=C_I \cap C_2 \cap C_3 \ldots\ldots \cap C_n$$

Where $\mu_c(y) = \min(\mu_c(\mu_c(y_1), \mu_c(y_2) \ldots.. \mu_c(y_n)))$

$\forall y \in Y$ where Y is the universe of discourse.

On the other hand, if the conclusion C to be drawn from a rule base R is the disjunction of the individual consequents of each rule then

$C_1 \cup C_2 \cup C_3 \ldots\ldots\ldots\ldots\ldots\ldots\ldots \cup C_n$

Where $\mu_c(y) = \max(_c(\mu_c(y_1), \mu_c(y_2) \ldots.. \mu_c(y_n))) \ \forall y \in Y$

Fuzzification

Fuzzification is the process of transforming crisp set to a fuzzy set or a fuzzy set to fuzzier set i. e. crisp quantities converted to fuzzy quantities. This operation translates crisp input values into linguistic variables. In the real-life world, the quantities may be thought of as crisp, accurate and deterministic, but actually they are not so. They possess uncertainty within themselves. The uncertainty may arise due to vagueness, imprecision and ambiguity. In this case the variable is probably fuzzy and can be represented by a membership function.

For example, when one is told that temperature is 9^0C, then person translates the crisp input value into linguistic variable such as "cold" or "warm" according ones knowledge and then make a decision about the need to wear jacket. If one fails to fuzzify then it is not possible to take decision or error decision may be reached.

For a fuzzy set $\tilde{A} = \{x_i, \mu_{\tilde{A}}(x_i): x_i \in X\}$ a common fuzzification algorithm is performed by keeping $\mu_{\tilde{A}}(x_i)$ constant and x_i being transformed into fuzzy set $\tilde{Q}(x_i)$ depicting the expression about x_i. The fuzzy set $\tilde{Q}(x_i)$ is kernel of fuzzification.

The fuzzified set $\tilde{A}$ be expressed as

$\tilde{A} = \mu_{\tilde{A}}(x_1).\tilde{Q}(x_1) + \mu_{\tilde{A}}(x_2).\tilde{Q}(x_2) + \ldots\ldots\ldots + \mu_{\tilde{A}}(x_n).\tilde{Q}(x_n)$ for I = 1 to n

This Process of fuzzification is called support (s-fuzzification) fuzzification. There is another method of fuzzification is called g-fuzzification or grade fuzzification. In this method x_i is kept constant and $\mu_{\tilde{A}}(x_i)$ expressed as a fuzzy set.

Defuzzification-

In many situations, for a system whose output is fuzzy, it is easier to take crisp decision if the output is represented as a single scalar quantity. This conversion of a fuzzy set to single crisp value is called defuzification and it the reverse process of fuzzification..
Methods for defuzzification are

(a) Centroid Method

Also known as the centre of gravity or centre of area method, It obtains the centre of area (x*) occupied by the fuzzy set. The expression for x* is given as

$$X^* = \frac{\int \mu(x)x\,dx}{\int \mu(x)dx}$$ for a continuous membership function,

and $$X^* = \frac{\sum_{i=1}^{n} x_i.\mu(x_i)}{\sum_{i=1}^{n} \mu(x_i)}$$ for a discrete membership function.

Here n represents the number of elements in the sample, x_i" are the elements, and μ (x_i) is its membership function.

(b)Centre of Sums (COS) Method-

In this method, overlapping area is counted twice whereas in centroid method overlapping area is counted once. COS builds the resultant membership function by taking the algebraic sum of outputs from each of the contributing fuzzy sets A_1, A_2, A_3, etc. The defuzzified value x* is given by

$$X^* = \frac{\sum_{i=1}^{N} x_i.\sum_{k=1}^{n} \mu_{\tilde{A}_k}(x_i)}{\sum_{i=1}^{N}.\sum_{k=1}^{n} \mu_{\tilde{A}_k}(x_i)}$$

Here ***n*** is the number of fuzzy sets and ***N*** is number of fuzzy variables. COS is actually the most commonly used defuzzification method. It can be implemented easily and leads to rather fast inference cycles.

(c) **Mean of Maxima (MOM) Defuzzification-**

One simple way of defuzzifying the output is to take the crisp value with the highest degree of membership. In cases with more than one element having the maximum value, the mean value of the maxima is taken. The equation for defuzzified value x* is given by

$$X^* = \frac{\sum_{x_i \epsilon M} x_i}{|M|}$$ Where **M= {x_i /μ (x_i)** equal to the height of fuzzy set}

|M| is the cardinality of set M. In the continuous case M, could be defined as

M={x€[c,-c] is equal to the height of fuzzy set}

In such case, the mean of maxima is the arithmetic average of mean value of all intervals contained in M including o length intervals.

The height of fuzzy set $\tilde{A}$ i.e h ($\tilde{A}$) is the largest membership grade obtained by any element in the $\widetilde{A}$ set.

Example on difuzzification

$\widetilde{A}_1$, $\widetilde{A}_2$, $\widetilde{A}_3$, are three fuzzy sets as shown below in figure 4.1, figure 4.2, and figure 4.3 respectively and aggregate is give in figure 4.4

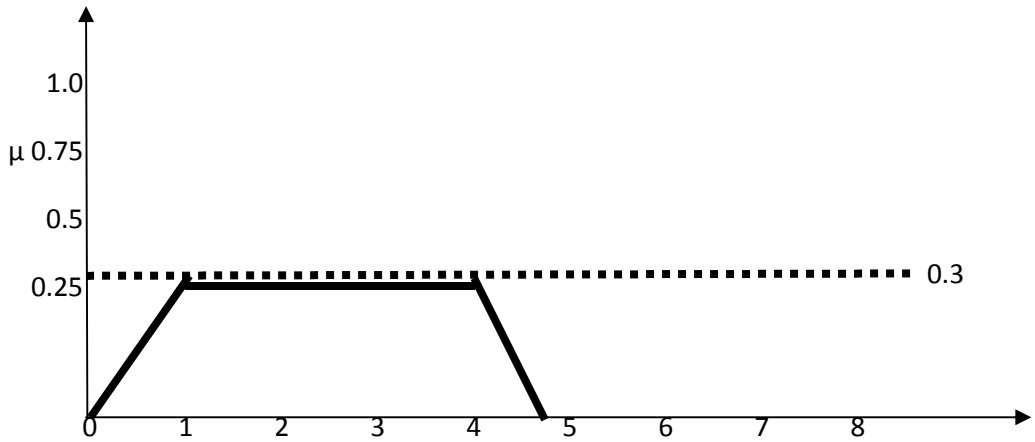


Figure 4.1 $\widetilde{A}_1$

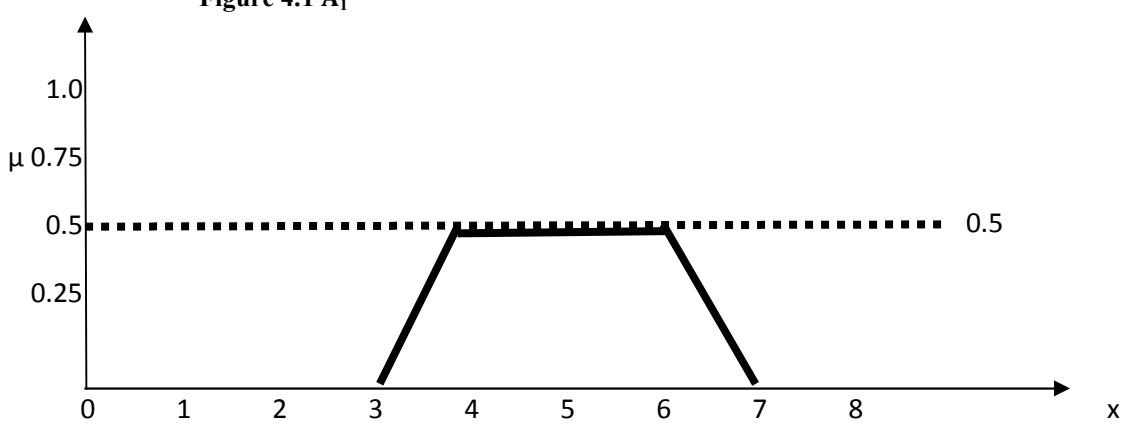


Figure 4.2 $\widetilde{A}_2$

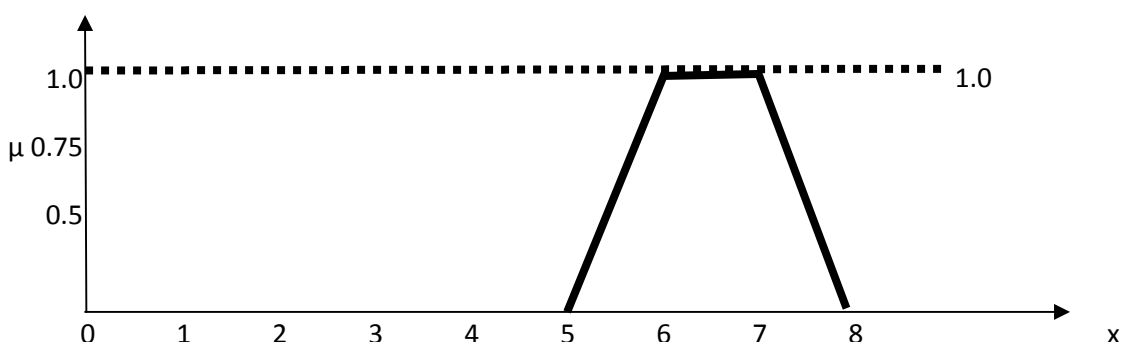


Figure 4.3 $\widetilde{A}_3$

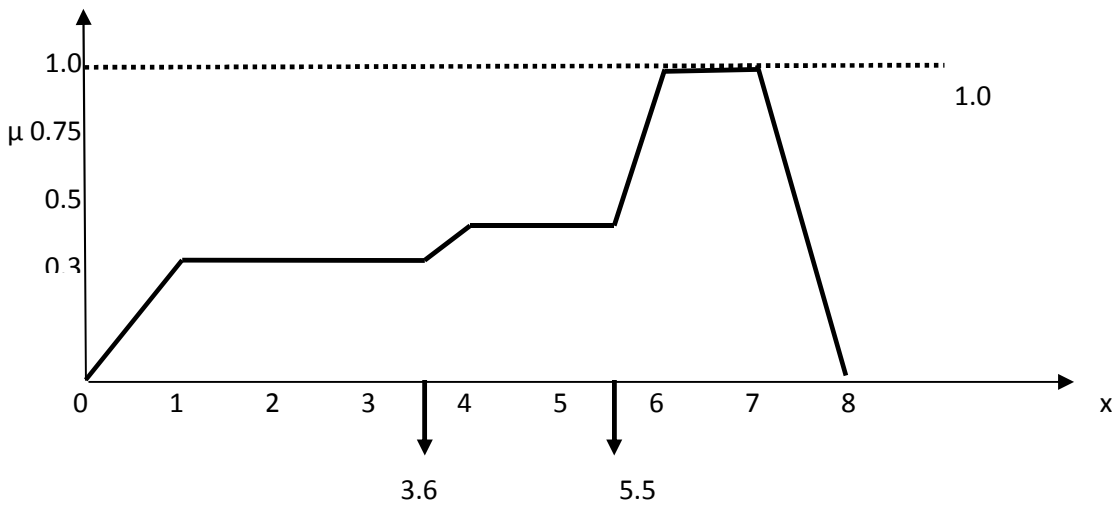


Figure 4.4 Aggregate of fuzzy sets Ã1, Ã2, Ã3

The defuzzification using (1) centroid method (2) COS method and (3) mean of maxima method is presented below

Centroid Method

To Compute X* the centroid , we view the aggregate fuzzy set as shon in figure. In figure the aggregate output has been divided into areas for better understanding

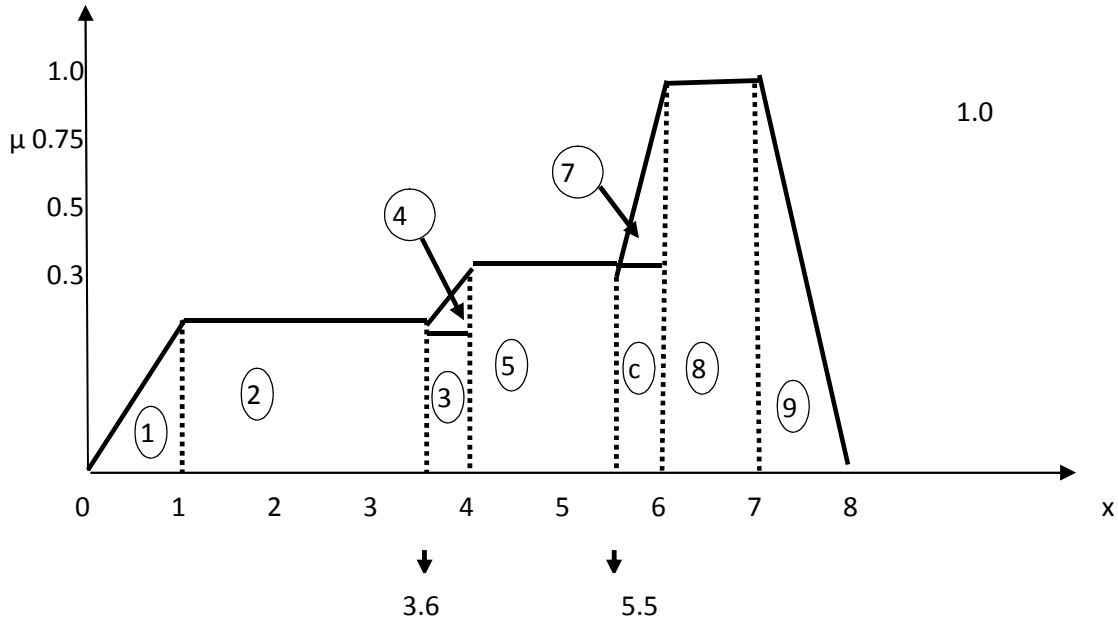


Figure 4.5 Aggregated fuzzy sets of Ã1, Ã2, Ã 3 viewed as area segments

For computation of x* we use table below

Area segment No	AREA (A)	$\overline{X}$	A.$\overline{X}$
1	$\frac{1}{2}$×0.3×1 =0.15	0.67	0.1005
2	2.6×0.3 =0.78	2.3	1.794
3	0.3×0.4 = 3.8	3.8	0.456
4	$\frac{1}{2}$×0.4×0.2 =0.04	3.8667	0.1546
5	1.5×0.5 = 0.75	4.75	3.5642
6	0.5×0.5 =0.25	5.75	1.4375
7	$\frac{1}{2}$×0.5×0.5 =0.125	5.833	0.729
8	1×1 =1	6.5	6.5
9	1/2×1×1 =0.5	7.333	3.665
	ΣA=3.715		ΣA. $\overline{X}$=18.353

Note-Compute area and centroid of each segment separately using mathematical formulae. In figure 4.5 regions 1, 4, 7 are triangular and remaining are rectangular.

(Centroid for triangular region $\overline{X}=\frac{x_1+x_2+x_3}{3}$

In the table area (A) shows the area of segments of the aggregate fuzzy set and $\overline{X}$ shows the corresponding centroid.

Now

$$X^*=\frac{\sum A.\overline{X}}{\sum A}=\frac{18.353}{3.715}=4.9$$

Centre of Sums (COS) Method

Here, unlike centroid method the overlapping area is counted not once but twice. Making use of aggregated fuzzy set shown the centre of sums, x* is given by

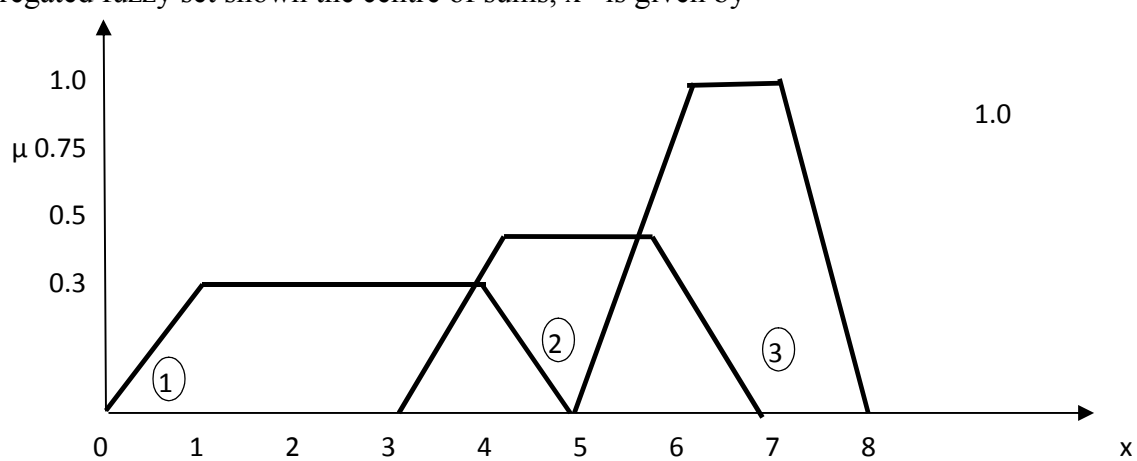


Figure 4.6 –***Aggregated fuzzy sets of $\tilde{A}1$, $\tilde{A}2$, $\tilde{A}$ viewed as area segments overlapping area counted twice***

For computation of x* we use table below

Area segment No	AREA (A)	$\overline{X}$	A.$\overline{X}$
1	$\frac{1}{2}\times0.3\times(3+5)=1.2$	2.5	3.00
2	$\frac{1}{2}\times0.5\times(4+2)=1.5$	5	7.5
3	$\frac{1}{2}\times2\times1\times(3+1)=2$	6.5	13.0
	ΣA=4.7		ΣA.$\overline{X}$ =23.5

$$X^* = \frac{\Sigma A.\overline{X}}{\Sigma A} = \frac{23.5}{4.7} = 5.0$$

Mean of Maxima (MOM) method

Since the aggregated fuzzy set shown in fig (d) is a continuous set x* the mean of maxima is computed as x*=6.5.

Here M = {X ∈[6, 7] /μ(x) =1] and the height of the aggregate fuzzy set is 1.

Figure 4.7 below shows the defuzzyfied outputs using above three methods

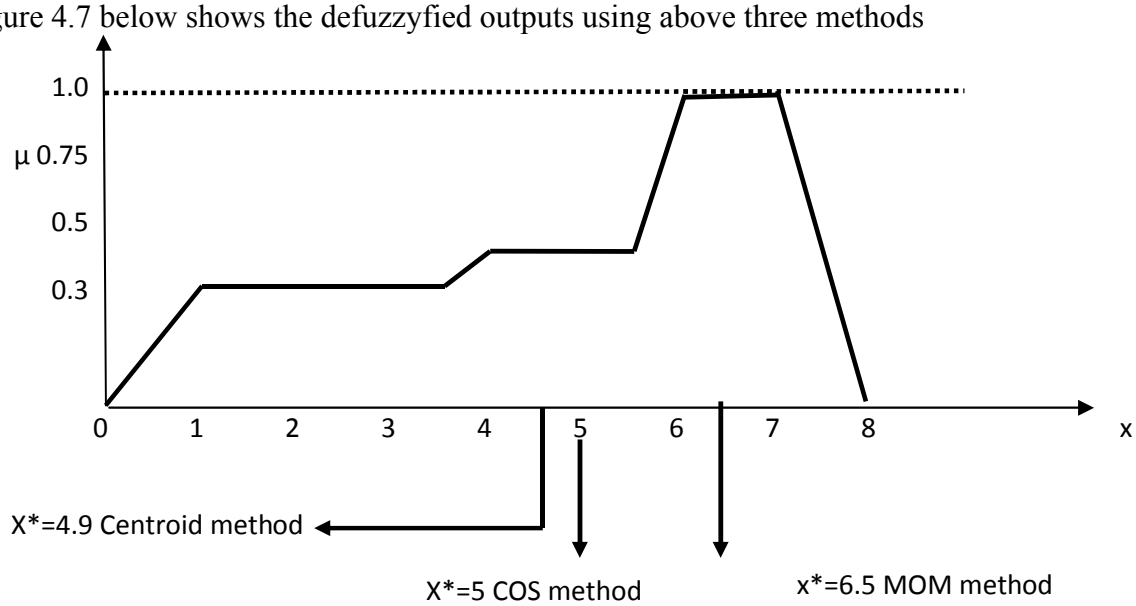


Figure 4.7 Defuzzified outputs of the sets of Ã1, Ã2, Ã3

Lecture # 5 Fuzzy Controllers, Industrial Applications –

The working of any fuzzy controller is shown in figure 5.1 below

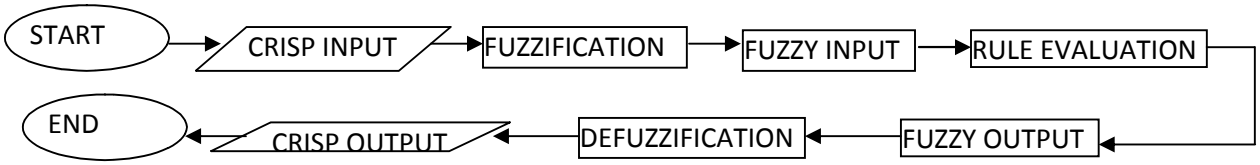


Figure 5.1 working of fuzzy controllers

Two fuzzy Controllers are

(i) Greg Viot's Fuzzy Cruise Control System
(ii) Yamakawa,s Air Conditioner Controller
(iii)

Greg Viot, s Fuuzzy Cruise Controller

This controller is used to maintain a vehicle at desired speed. The system consists of two fuzzy inputs, namely speed difference speed difference and acceleration, one fuzzy output, namely throttle control as illustrated in figure 5.2

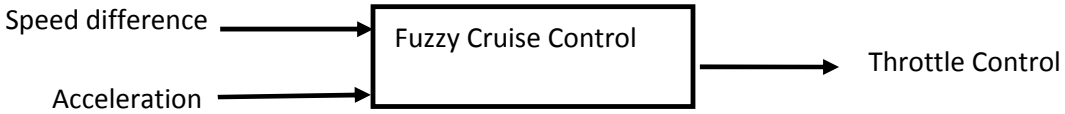


Figure 5.2 Fuzzy Cruise Control System

Fuzzy Rule Base

A sample fuzzy rule base R governing the fuzzy cruise Control is as given in table below

Table-1 Fuzzy Rule Base for Cruse Control

Rule 1	*If(speed difference is NL) and (acceleration is ZE) then (throttle control is PL)*
Rule 2	*If(speed difference is ZE) and (acceleration is NL) then (throttle control is PL)*
Rule 3	*If(speed difference is NM) and (acceleration is ZE) then (throttle control is PL)*
Rule 4	*If(speed difference is NS) and (acceleration is ZE) then (throttle control is PL)*
Rule 5	*If(speed difference is PS) and (acceleration is ZE) then (throttle control is PL)*

Rule 6	*If(speed difference is PL) and (acceleration is ZE) then (throttle control is PL)*
Rule 7	*If(speed difference is ZE) and (acceleration is ZE) then (throttle control is PL)*
Rule 8	*If(speed difference is ZE) and (acceleration is ZE) then (throttle control is PL)*

Keys
NL – Negative Large NM – Negative Medium, NS - Negative Small, ZE- - Zero
PS – Positive Small, PM - Positive Medium, PL- Positive Large

Fuzzy Sets

The fuzzy sets which characterize the inputs and outputs are as given in figures

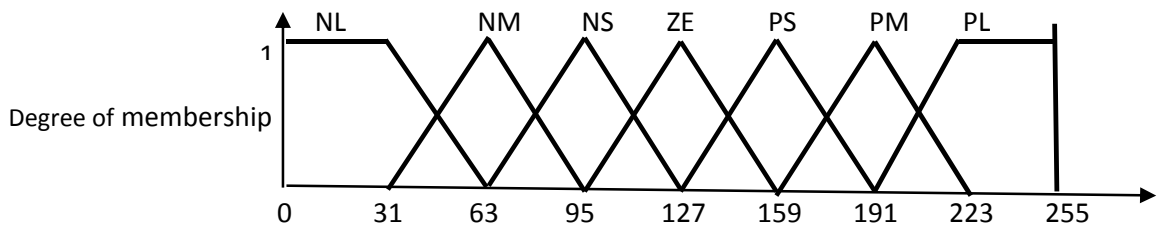


Figure 5.3 Speed difference (Normalized)

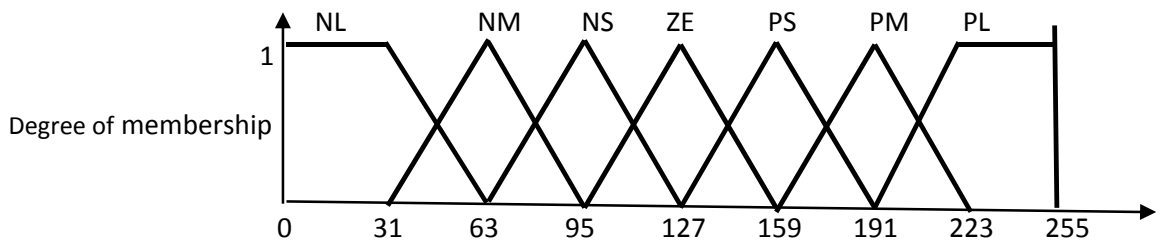


Figure 5.4 Acceleration (Normalized)

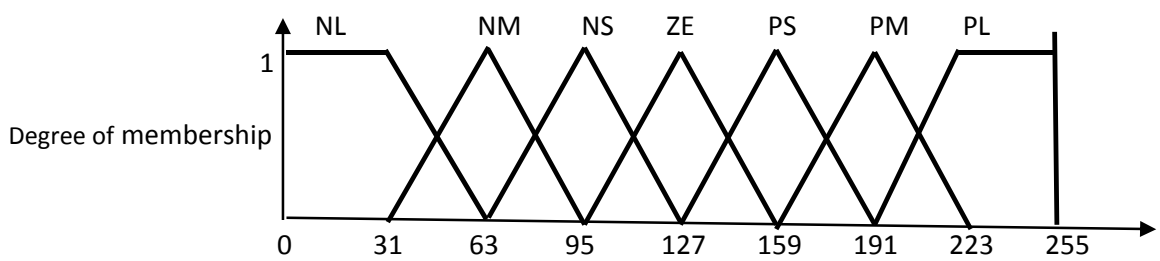
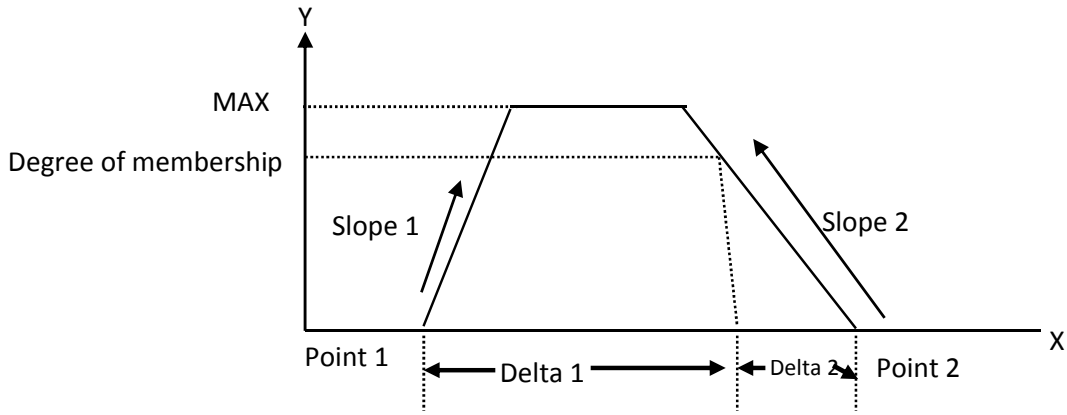


Figure 5.5 Throttle control (Normalized)

Fuzzification of Inputs

For the fuzzification of inputs, that is, to compute the membership for the antecedents, the formula illustrated in Figure 5.6) is used.

Degree of membership



- Compute
  Delta 1=x-point1
  and
  Delta 2 =point2-x

Figure 5.6 - Computing of fuzzy membership values

If (Delta 1≤0) or (Delta2≤0)
Then degree of membership = 0

Else degree of membership=min $\begin{bmatrix} Delta\ 1 * Slope1 \\ Delta\ 2\ *\ Slope\ 2 \\ MAX \end{bmatrix}$

Here, x which is the system input has its membership function value computed for all fuzzy sets. For example, the system input speed difference deal with 7 fuzzy sets, namely NL, NM, NS, ZE, PS, PM, PL. For a measured value of speed difference x', the membership function of x, in each of the seven sets in computed using formula shown in previous figure (l). Let μ_1 , μ_2 , μ_3 , μ_4 , μ_5 , μ_6 , μ_7 , , be the seven membership values. Then all values are recorded for the input x' in an appropriate data structure.

Similarly, for each of the other system inputs (acceleration in this case), the fuzzy membership value are recorded.

Example

Let the measured normalized speed difference be ***100*** and the normalized acceleration is ***70***, then the fuzzified inputs after computation of the fuzzy membership values are shown in figure 5.7 below.

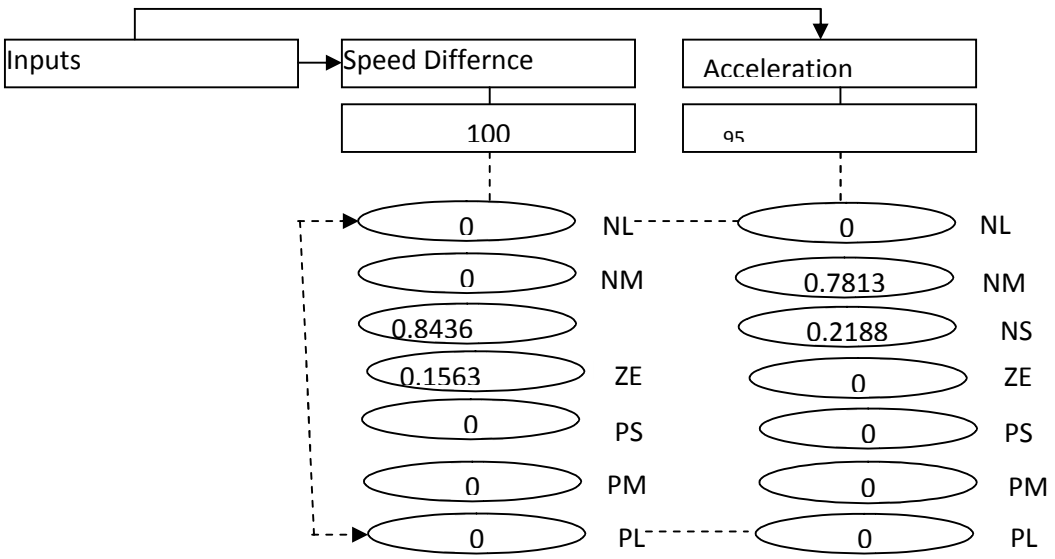


Figure (m)5.7 Recorded values for seven different sets for speed difference =100 and Acceleration=70

The computations of the fuzzy membership values for the given inputs have been shown n figure 5.8 below

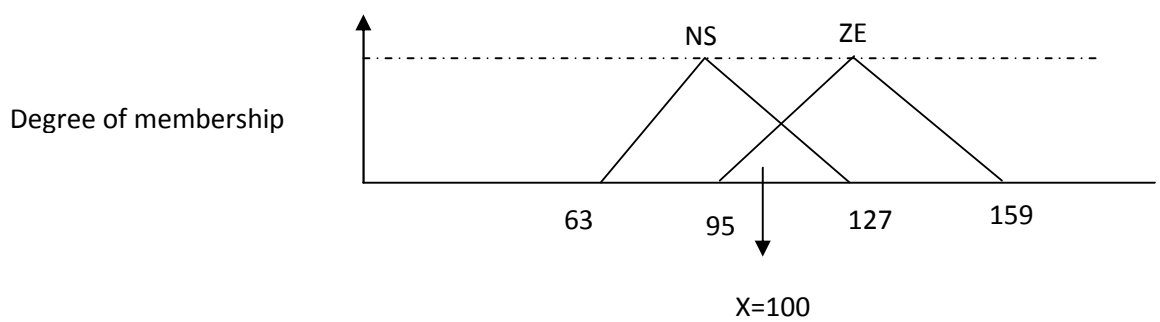


Figure 5.8 Fuzzification of speed difference = 100

For the speed difference (x-=100) the qualifying sets are shown figure. Fuzzy membership function of x for NS where

Delta 1 100-63=37
Delta 2 127-100=27
Slope 1=1/32=0.03125
Slope 2 =1/32=0.03125

Degree of membership function $\mu_{NS}(x)$ $=\min\begin{pmatrix} 37 \times 0.03125 \\ 27 \times 0.03125 \\ MAX \end{pmatrix}$ =0.8438

Fuzzy membership function of x for ZE where

Delta 1 100-95=5
Delta 2 159-100=59
Slope 1=1/32=0.03125
Slope 2 =1/32=0.03125

Degree of membership function $\mu_{ZE}(x) = \min \begin{bmatrix} 5 \times 0.03125 \\ 59 \times 0.01325 \\ 1 \end{bmatrix} = 0.1563$

The membership function of x with the remaining fuzzy sets, namely NL, M, PS, PM, PL is zero. Similarly for acceleration(x=70) the qualifying fuzzy sets are shown in figure 5.9

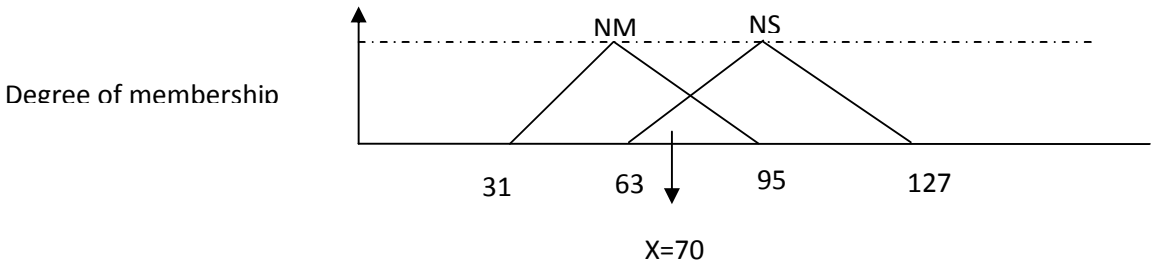


Fig 5.9 Fuzzification of Acceleration

The fuzzy membership function of x=70 for NM is μ_{NM} =0.7813 and for NS is μ_{NS} =0.2188 and for remaining sets membership value will be zero.

Rule Strength Computation

The rule strength is obtained by computing the minimum of the membership function of the antecedents.

Example

For the same rule base R given in table, the rule strength using the fuzzy membership value illustrated in figure

Rule 1: min (0, 0) =0

Rule 2: min (0.1563, 0) =0

Rule 3: min (0, 0) =0

Rule 4: min (0.8438, 0) =0

Rule 5: min (0, 0.2188) =0

Rule 6: min (0, 0) =0

Rule 7: min (0.1563, 0.2188) =0.1563

Rule 8: min (0.1563, 0.7813) =0.1563

(For strength computation use rule from rule base R in <u>Table 1</u> and for each rule put corresponding membership values for speed difference and acceleration and then find minimum for example

Rule 1	**If(speed difference is NL) and (acceleration is ZE) then (throttle control is PL)**

Here Speed difference is NL as we computed early $\mu_{NL}=0$, Acceleration is ZE as we computed early $\mu_{ZE}=0$

min ($\mu_{NL}=0$, $\mu_{ZE}=0$) =min (0, 0) =0)

The fuzzy output of the system is the 'fuzzy OR' of all fuzzy outputs of the rules with non-zero strengths. In the event of more than one rule qualifying for the same fuzzy output, the stronger among them is chosen.

Example

In the given rule base R, the competing fuzzy outputs are those of Rule 7 and 8 with strengths of each are 0.1563. However, the fuzzy outputs computed here do not aid clear-cut decision on the throttle control. Hence the need for defuzzyfication arises.

Defuzzification

The centre of gravity method is applied to defuzzify the output. Initially the centroid is computed for each of computing output membership function. Then, the new output membership areas are determined by shortening the height of the membership value on the Y axis as dictated by the rule strength value. Finally, the Centre of Gravity (CG) is computed using the weighted average of the X-axis centroid points with the newly computed outputs area, the letter serving as weights.

Example

Figure 5.10 illustrates the computation of CG for the two competing outputs of rule 7 and 8 with strength of 0.1563 each

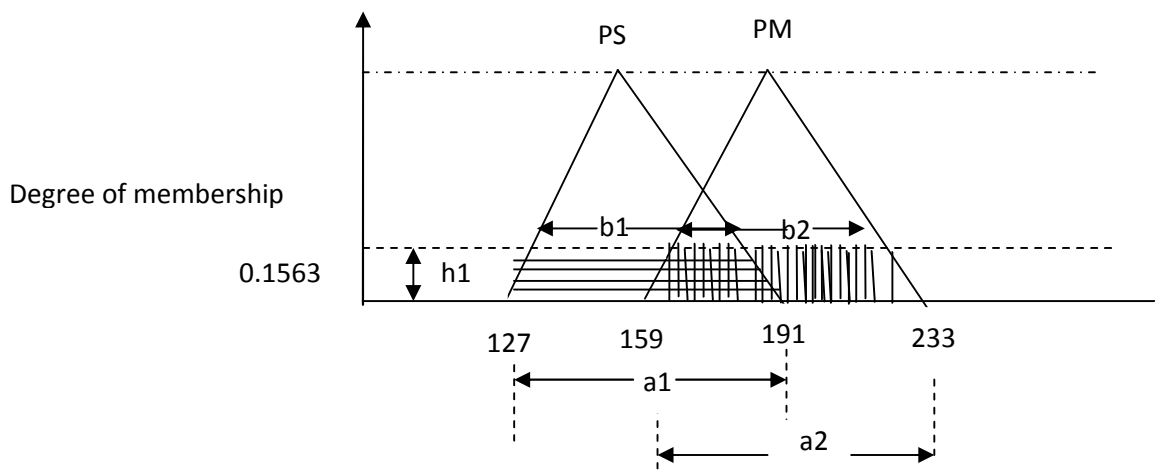


Figure 5.10) Computation of entre of Gravity

For the fuzzy set PS

X-axis centroid	=159
Rule strength applied to determine output area	=0.1563
Shaded area =×h× (a1+b1) = $\frac{1}{2}$(0.1563) × (64+63.82)	=9.99

For the fuzzy set PM

X-axis centroid point	=191
Rule strength applied to determine output area	=0.1563
Shaded area =$\frac{1}{2}$.h. (a1+b1) =$\frac{1}{2}$. (0.1563)(64+63.82)	=9.99

Therefore

$$\text{Weighted average (CG)} = \frac{9.99\times159+9.99\times191}{9.99+9.99} = 175$$

In crisp terms the throttle control (normalized) is to be set as 175.